

LINEAR ALGEBRA: MATRIX TRANSFORMATIONS

IDS + PROPERTIES

$$\begin{aligned}(AB)^T &= B^T A^T & \det(A^T) &= \det(A) \\ (A^T)^T &= (A)^T & \det(cA) &= c^n \det(A) \\ (A+B)^T &= A^T + B^T & \det(AB) &= \det(A)\det(B) \\ (AB)^T &= B^T A^T & \det(A^T) &= \frac{1}{\det(A)} \\ & & \det(A^n) &= \det(A)^n\end{aligned}$$

rank: # of linearly independent rows or columns (both=)

$$\begin{aligned}\text{rank}(A+B) &\leq \text{rank}(A) + \text{rank}(B) \\ \text{rank}(A) &= \text{rank}(A^T) = \text{rank}(A^T A) = \text{rank}(A A^T)\end{aligned}$$

MATRIX TRANSPOSE

$$\begin{pmatrix} a_1 & b_1 & c_1 & \dots & x_1 \\ a_2 & & & & \\ \vdots & & & & \\ a_n & & & & x_n \end{pmatrix}^T = \begin{pmatrix} a_1 & a_2 & \dots & a_n \\ b_1 & & & \\ \vdots & & & \\ x_1 & \dots & x_n \end{pmatrix}$$

PROJECTIONS

$\hat{p}$: projection of $\vec{v}_2$ onto $\vec{v}_1$

$$\begin{aligned}h &= (\vec{v}_1 \cdot \vec{v}_1)(\vec{v}_1 \cdot \vec{v}_2) - (\vec{v}_2 \cdot \vec{v}_2) \\ &= a^2 d^2 + b^2 c^2 - 2abcd\end{aligned}$$



NOTES

IF $A \in \mathbb{R}^{m \times m}$ has linearly dependent cols, then projection onto $R(A)$ is

$$P = A(A^T A)^{-1} A^T$$

DETERMINANTS

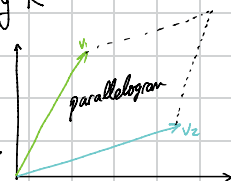
2x2 Matrix

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Other rules:

- $n \times n$ determinant: chain the top row with det below, sm. to 3×3
- scale 1 row by K : scale det by K
- scale n rows by K : scale det by K^n
- Det doesn't change after row operations
- When 2 identical rows, $\det = 0$

Determinant can be considered the "area" of its vectors



- IFF $\det \neq 0$, precisely one solution to system. otherwise ($\det = 0$), system is linearly dependent.

MATRIX INVERSE

$$[A | I] = [I | A^{-1}]$$

To Find inverse for A (A^{-1})

- Identify row operations for $A \rightarrow I$
- Perform those operations on I (i.e. find the rref)

NOTES

$$2 \times 2 \text{ inverse: } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- Matrices with $\det = 0$ are linearly dependent and not invertible
- Non-square matrices are not invertible
- $AA^{-1} = A^{-1}A = I_n$